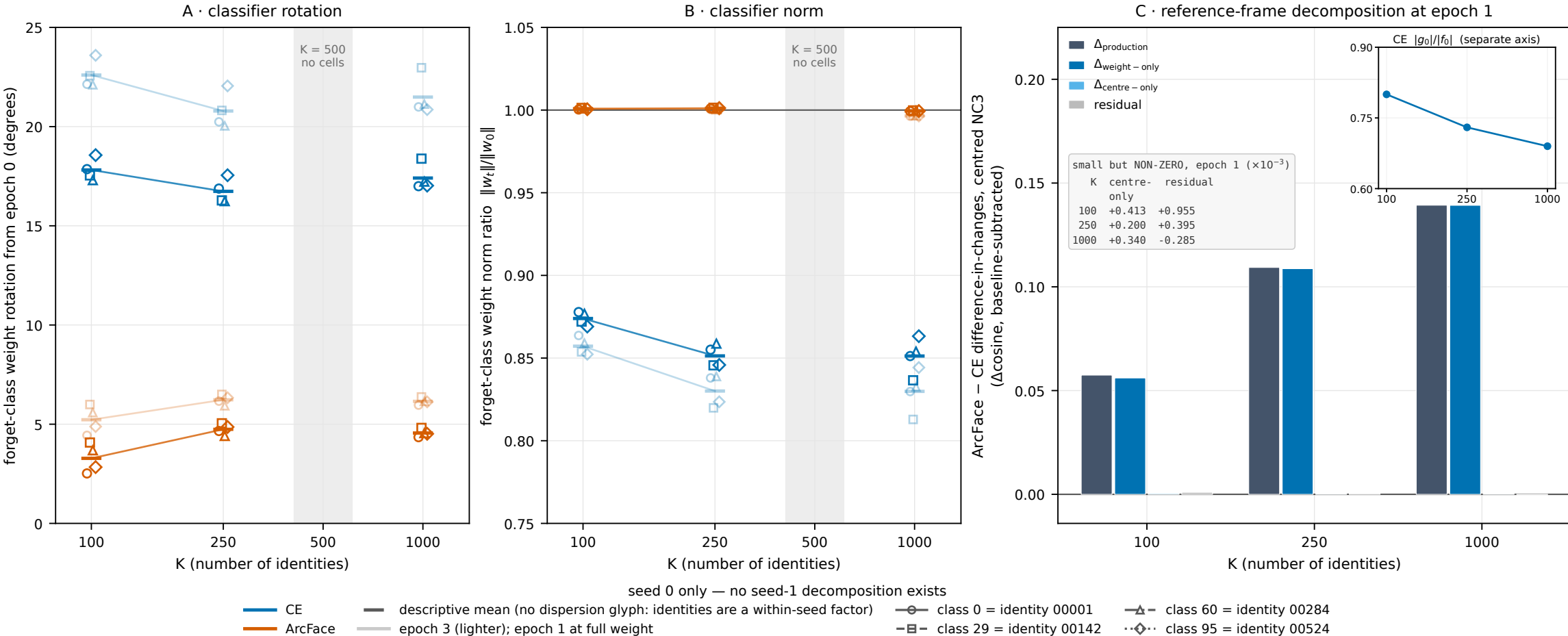




Panel C. The residual is production minus weight-only minus centre-only, exactly, per cell. The four corners are an arithmetic decomposition of the metric $A(w, c) = \cos(f_0 - g_0, w - c)$, not an intervention on training, and not a measure of unlearning quality.

Panel B. Descriptive. A cosine is invariant to scaling its whole argument, so the uncentred cosine is unaffected by the weight norm; the centred cosine takes $w - c$, and changing w against a fixed c can change that direction. No attribution from these ratios is claimed.

K = 500 carries no unlearning cells. Baselines: CE 68.9676 %, ArcFace 71.3232 %, absolute gap 2.3556 pp, exceeding the prespecified 2.00 pp head-fairness gate. The gate is symmetric and failed because ArcFace EXCEEDED CE, not because ArcFace underperformed. Its cells were never run and neither head was returned to rescue it; both baselines are retained as artifacts.

That is a deliberate exclusion under a prespecified rule, not missing data — no value is imputed and no line is drawn across the gap in any panel.